GALGOTIAS UNIVERSITY

SCHOOL OF COMPUTING SCIENCE AND ENGINEERING

Department of Computer Science and Engineering

PROJECT APPROVAL FORM AND ABSTRACT (ODD SEMESTER 2026-2027)

1. PROJECT APPROVAL FORM DETAILS

Project Title	**ANVESH:** An Explainable Multi-Tier AI Architecture and Forensic Signal Fusion Pipeline for Advanced Email Threat Attribution and Chain-of-Custody Intelligence
Project Type	**AI/ML/Data Science**
Project Outcome	**Project and Research Paper**
SDG Goal Number	**SDG 16: Peace, Justice, and Strong Institutions** Target 16.4: Combating illicit cyber financial flows and cybercrime; Target 16.a: Strengthening institutions with forensic-grade evidentiary integrity and transparent attribution safeguards.
Publication Target	**SCOPUS Conference** (e.g., IEEE / Springer International Conference)
Guide Name	[To be filled by Student / Assigned Faculty Guide]
Area / Domain	**Cyber Forensics, Applied Machine Learning, and Information Security**

2. STUDENT DETAILS

Member	Student Name	Enrollment No.	Admission No.	Program / Branch	Semester
Student 1	[Student 1 Name]	[Enrollment No.]	[Admission No.]	B.Tech CSE	7th / 8th
Student 2	[Student 2 Name]	[Enrollment No.]	[Admission No.]	B.Tech CSE	7th / 8th

3. ABSTRACT (One Continuous Paragraph · 200–250 Words)

Rapid escalation in sophisticated email-based cyber threats, including Business Email Compromise (BEC), lookalike domain typosquatting, and multi-hop proxy obfuscation, presents critical challenges for cybersecurity operations and digital forensics. Conventional signature-based filters and monolithic black-box machine learning classifiers frequently suffer from high false-positive rates, inability to interpret obfuscated RFC-822 routing headers, and a critical lack of explainability necessary for court-admissible forensic evidence. To address these vulnerabilities, this project proposes ANVESH, an explainable multi-tier cyber forensic intelligence platform designed for automated email threat detection, cryptographic verification, and geolocation attribution. The proposed methodology integrates a modular machine learning pipeline consisting of calibrated natural language processing for phishing lexicon identification, financial coercion heuristics for BEC detection, and homoglyph distance metrics for typosquatting discovery. These analytical signals are synthesized through a forensic signal fusion engine that cross-correlates cryptographic authentication protocols (SPF, DKIM, DMARC) with authoritative Border Gateway Protocol (BGP) routing and anonymization telemetry. Furthermore, the system incorporates an append-only digital ledger with SHA-256 evidence hashing to guarantee an uncompromised chain of custody and automated generation of forensic dossiers. The expected outcome is a functional, full-stack intelligence platform that achieves high classification accuracy while providing transparent, human-in-the-loop explainability for incident response analysts. By preventing premature attribution and preserving evidentiary integrity, this research significantly fortifies enterprise defenses, strengthens law enforcement investigative capabilities, and advances trustworthy digital forensic science.

4. PROBLEM STATEMENT

Modern email-borne threats exploit advanced deception vectors—such as executive display-name spoofing, deceptive homoglyphs, and anonymized network relays (e.g., Tor and bulletproof residential proxies)—resulting in massive financial fraud and enterprise data compromises. Existing security systems operate predominantly as opaque, binary filters that fail to provide granular diagnostic rationales or forensic traceability across RFC-822 transport headers. Furthermore, investigative teams face severe attribution dilemmas, often conflating transit infrastructure with threat actor identity or compromising the legal chain of custody during evidence handling. There is a critical requirement for an open, explainable, and multi-layered forensic intelligence framework capable of deconstructing complex email headers, validating cryptographic authenticity, contextualizing threats via multi-model machine learning, and enforcing verifiable evidentiary neutrality.

5. PROPOSED METHODOLOGY & ARCHITECTURAL SOLUTION

The proposed solution, **ANVESH**, transforms raw email evidence (.eml, .msg) into actionable, legally verifiable forensic intelligence. It decouples threat detection into a specialized 4-tier pipeline: (1) Calibrated NLP classification for credential harvesting and deceptive intent; (2) Rule-assisted NLP for BEC and financial coercion syntax; (3) Levenshtein and Jaro-Winkler homoglyph metrics for lookalike domain discovery; and (4) An explainable forensic signal fusion engine that cross-correlates machine learning signals with SPF/DKIM/DMARC cryptographic verdicts and authoritative BGP autonomous system (ASN) telemetry. The platform guarantees strict chain-of-custody tracking via SHA-256 hashing, an append-only audit trail, and an evidentiary attribution safeguard ensuring transit networks are never conflated with physical human identities.

6. TECHNOLOGY STACK & IMPLEMENTATION ARCHITECTURE

Component	Selected Technologies & Frameworks	Academic / Engineering Justification
Languages	Python 3.11+, TypeScript, JavaScript, SQL	High performance data parsing, robust typing, and broad ML ecosystem.
Backend Framework	FastAPI (Async REST API, Pydantic v2)	Sub-millisecond latency, automatic OpenAPI docs, and async I/O.
Frontend / SOC UI	React 18, Vite, Tailwind CSS, Lucide Icons	High-density SOC analyst dashboard with interactive campaign graph.
Mobile Companion	React Native (Expo SDK 51)	On-call incident responder alert triage and rapid push review.
AI & Machine Learning	Scikit-Learn, TF-IDF Vectorizer, Calibrated Logistic Regression, Jaro-Winkler	Lightweight, deterministic, explainable inference suited for real-time triage.
Forensic & Network	Python email/MIME parser, DNSSEC/SPF/DKIM/DMARC validators, IPwhois BGP ASN, Tor Node Intel	RFC-822 standard compliance and zero-fabrication perimeter extraction.
Database & Security	PostgreSQL (Supabase) + SHA-256 evidence ledger	Append-only audit trail guaranteeing tamper-evident chain of custody.
Dossier Generation	ReportLab PDF Generator	One-click automated generation of court-ready forensic reports.

7. KEY INNOVATIONS, RESEARCH NOVELTY & EXPECTED OUTCOME

Core System Features	• **RFC-822 Parsing & Hop Traversal:** Traverses multi-hop Received headers from perimeter back to origin IP with RFC-1918 filtering. • **Cryptographic Matrix:** Evaluates SPF, DKIM, and DMARC to distinguish misconfigurations from active header tampering. • **Multi-Model AI Pipeline:** Decoupled lightweight classifiers for phishing, BEC financial coercion, and display-name spoofing. • **Homoglyph Detection:** Algorithmic typosquatting identification using edit-distance thresholds against known brands. • **Court-Ready Dossier:** One-click export of formal PDF investigation dossiers with SHA-256 evidence fingerprinting.
Research Novelty	1. **Explainable Forensic Signal Fusion vs. Black-Box Models:** Fuses discrete cryptographic checks, BGP telemetry, and lightweight ML inferences into an interpretable score (0–100) auditable in legal proceedings. 2. **Ethical Attribution Invariant:** Implements an architectural safeguard preventing false attribution by strictly separating transport network routing from human threat actor identity. 3. **Multi-Vector Discrepancy Matching:** Combines header discrepancies (From vs. Reply-To) with semantic NLP urgency analysis to capture zero-day attacks that bypass standard filters.
Expected Outcome	• A fully functional, production-ready web platform and companion mobile triage app processing RFC-822 email evidence in < 2s. • An explainable forensic assessment engine achieving high classification accuracy (>95%) across benchmark datasets. • A publishable research paper targeting a **SCOPUS Conference** (IEEE/Springer) documenting the fusion methodology.

Student 1 Signature	Student 2 Signature	Faculty Guide Signature	Project Coordinator / HOD